

0.1 Section 4.3 - Primes

Definition 1

A **prime number** is a positive integer with exactly 2 **distinct factors**

Example 1

Here are the first few primes

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, ...

is 257 prime?

We use **trial division**

We'll use a loop

```
for k = 2 to 256
  if 257 % k = 0
    then return "composite"
  else
    k += 1
```

For efficiency, we could just go up to $\sqrt{257}$, or $\lfloor \sqrt{256} \rfloor$

Second method: **Sieve of Eratostheres**

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50

I honestly forgot how this works since I didn't write it down too well

Third method: **Fermat's Method**

if $n = a^2 - b^2$

then $n = (a - b)(a + b)$

Check: $a - b > 1$ and $a + b > 1$

If neither is 1, then n is composite If both are greater than 1, then they can't be n

Fourth Method: **Quadratic Sieve**

Solve

$$a^2 - b^2 \equiv 0 \pmod{n}$$

Then

$$n \mid a^2 - b^2 \implies a^2 - b^2 = nk, k \in \mathbb{Z}$$

Proposition 1

There are infinitely primes

Proof

Suppose on the contrary, there are finitely many primes such that all primes can be written like

$$p_1, p_2, \dots, p_k$$

Consider

$$N = p_1 p_2 \cdots p_k + 1$$

Case 1: Suppose N is prime

That means that

N is larger than p_i for $i = 1, 2, \dots, k$

Hence $N \neq p_i$ for $i = 1, 2, \dots, k$ We have a contradiction of our finitely many primes!

Case 2: N is composite

Then N must have a prime factor in the list $p_1, p_2, \dots, p_k$

Suppose p_j is a prime factor of N

Then $p_j \mid N$ and not $p_j \mid p_1 p_2 \cdots p_k$

Then we conclude

$$p_j \mid (N - p_1 p_2 \cdots p_k)$$

Therefore

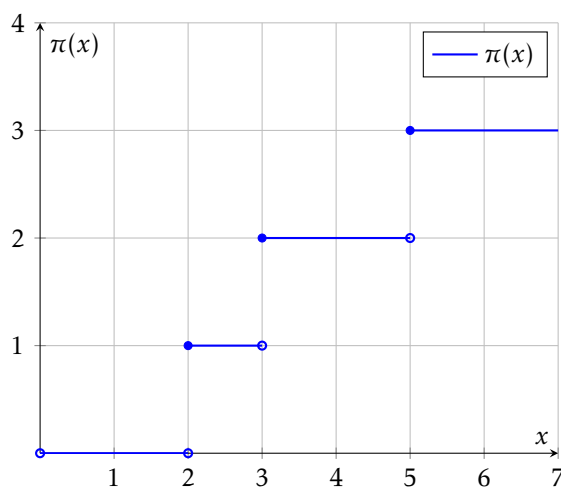
$$p_j \mid 1, \text{ is a contradiction}$$

We conclude there are infinitely primes, as desired (by contradiction)

Definition 2 (Prime Counting Function (π))

The function $\pi : \mathbb{R}^+ \rightarrow \mathbb{Z}^+$ is defined by

$$\pi(x) = \text{number of primes less than or equal to } x$$



Theorem 1 (The Prime Number Theorem)

This theorem is defined that $\pi(x)$ is approximately equal to $\frac{x}{\ln(x)}$ as $x \rightarrow \infty$

$$\lim_{x \rightarrow \infty} \frac{\pi(x)}{\frac{x}{\ln(x)}} = 1$$

In other words,

$$\pi(x) \approx \frac{x}{\ln(x)}$$

Note

This was proven independently by Jacques Hadamard and Charles Jean de la Vallée Poussin in 1896

To recap, we know so far for $a, b, c, d \in \mathbb{Z}, m \in \mathbb{Z}^+$ that

1. $a \equiv b \pmod{m} \iff m \mid (a - b)$
2. $a \equiv b \pmod{m}, c \equiv d \pmod{m}$
 - (i) $a + c \equiv b + d \pmod{m}$
 - (ii) $ac \equiv bd \pmod{m}$
 - (iii) $ac \equiv bc \pmod{m}$

But we see that we can multiply and add with $\equiv$, but how can we divide with modular arithmetic?

Let's go back to algebra

How do we solve this? We need a way to get rid of the 7

$$\begin{aligned}7x &= 14 \\ \frac{7}{7}x &= \frac{14}{7} \\ 1 \cdot x &= 2 \\ \boxed{x = 2}\end{aligned}$$

In this case, we used the **multiplicative inverse** of 7, 7^{-1} , which is $\frac{1}{7}$

$$\begin{aligned}7x &= 14 \\ 7^{-1}7x &= 7^{-1} \cdot 14 && \text{(Multiplicative Inverse)} \\ 1x &= 2 && \text{(Multiplicative Identity)} \\ x &= 2\end{aligned}$$

In order to divide, mod m , we need to understand **multiplicative inverse**

1 is still the multiplicative identity, so given $a \in \mathbb{Z}$, the multiplicative inverse of a (which we'll call x) satisfies

$$ax \equiv 1 \pmod{m}$$

We conclude that

$$m \mid (ax - 1)$$

This means that

$$\begin{aligned}\exists k \in \mathbb{Z} \text{ such that} \\ ax - 1 &= mk \\ \text{and} \\ ax - mk &= 1\end{aligned}$$

Our first task is: when does this equation have a solution?

Claim 1

If $m \mid a$ then $ax - mk = 1$ has no integer solutions

Proof

If $m \mid a$ then $\exists n \in \mathbb{Z}$ such that $a = nm$

The equation $ax - mk = 1$ becomes

$$ax - mk = 1$$

$$nm - mk = 1$$

(Substitute)

$$m(n - k) = 1$$

(Factor m)

We conclude that $m \mid 1$, which is not possible

So $ax - mk = 1$ has no integer solutions

Example 2

Let's do an example

Let $m = 12, a = 7$

We want to find the multiplicative inverse of 7 modulo 12

We must solve

$$7x \equiv 1 \pmod{12}$$

Method 1: Brute Force

$$7 \cdot 1 \pmod{12} = 7$$

$$7 \cdot 2 \pmod{12} = 2$$

$$7 \cdot 3 \pmod{12} = 9$$

$$7 \cdot 4 \pmod{12} = 4$$

$$7 \cdot 5 \pmod{12} = 11$$

$$7 \cdot 6 \pmod{12} = 6$$

$$7 \cdot 7 \pmod{12} = 1$$

$7^{-1} \equiv 7 \pmod{12}$

This wasn't too long, but it can be very tedious and very long as we work with higher numbers. This works, but it's not good.

Method 2: Adding 0

Since $0 \equiv 12 \pmod{12}$, we can keep on adding 12 to the right side until it's divisible by 7

$$\begin{array}{r}
 7x \equiv 1 \pmod{12} \\
 +0 \equiv 48 \pmod{12} \\
 \hline
 7x \equiv 49 \pmod{12} \\
 7^{-1}7x \equiv 7^{-1}7 \cdot 7 \pmod{12} \\
 \boxed{x \equiv 7 \pmod{12}}
 \end{array}$$

This was pretty short, but we got lucky. Normally this would take a lot more steps / additions

Method 3: Euclidean Algorithm

This is the one we'll be using most in this class

Let's rewrite

$$7x \equiv 1 \pmod{12}$$

We can rewrite it as divides

$$12 \mid (7x - 1)$$

Since 12 divides $(7x - 1)$, that means $(7x - 1)$ is a multiple of 12, so

$$\begin{aligned}
 7x - 1 &= 12k \\
 7x - 12k &= 1
 \end{aligned}$$

Now we just repeat the **division algorithm**

We start by breaking up 12

$$\begin{aligned}
 12 &= 1 \cdot 7 + 5 \\
 7 &= 1 \cdot 5 + 2 \\
 5 &= 2 \cdot 2 + 1
 \end{aligned}$$

Once we have the remainder as 1, we must work backwards

From the equations above, let's solve for each remainder

$$5 = 2 \cdot 2 + 1$$

(Let's solve for 1)

$$1 = 5 - 2 \cdot 2$$

$$7 = 1 \cdot 5 + 2$$

(Solve for 2)

$$2 = 7 - 1 \cdot 5$$

$$12 = 1 \cdot 7 + 5$$

(Solve for 5)

$$5 = 12 - 1 \cdot 7$$

Now, using $1 = 5 - 2 \cdot 2$ as our base, let's repeatedly substitute the remainders

For each line, substitute the divisor then simplify

$$1 = 5 - 2 \cdot 2$$

(now substitute 2)

$$1 = 5 - 2(7 - 1 \cdot 5)$$

(now simplify)

$$1 = 5 - 2 \cdot 7 + 2 \cdot 5$$

$$1 = 3 \cdot 5 - 2 \cdot 7$$

(now substitute 5)

$$1 = 3(12 - 1 \cdot 7) - 2 \cdot 7$$

(now simplify)

$$1 = 3 \cdot 12 - 3 \cdot 7 - 2 \cdot 7$$

$$1 = 3 \cdot 12 - 5 \cdot 7$$

Remember that we were trying to solve for $7x - 12k = 1$, so we conclude that

$$7x - 12k = 1$$

$$7(-5) - 12(-3) = 1$$

$$x = -5, k = 3$$

We conclude that

$$\begin{array}{l} x \equiv -5 \pmod{12} \\ +0 \equiv 12 \pmod{12} \end{array}$$

$$x \equiv 7 \pmod{12}$$

0.1.1 Euclidean Algorithm

Definition 3 (Euclidean Algorithm)

Because that last example could be hard to read, I'll rewrite the Euclidean Algorithm but with variables

Let's say we're trying to find the multiplicative inverse of

$$a \pmod{m}$$

We essentially solve

$$a \cdot a^{-1} - mk = 1, \text{ k doesn't matter}$$

Using the Euclidean Algorithm, solve the following until the remainder becomes 1

If the remainder equals 0, that means that the number does not have a multiplicative inverse mod m

$$\begin{array}{ll} m = k_1 \cdot a + r_1 & \text{(first iteration)} \\ a = k_2 \cdot r_1 + r_2 & \text{(second iteration)} \\ r_1 = k_3 \cdot r_2 + r_3 & \text{(third iteration)} \\ r_2 = k_4 \cdot r_3 + 1 & \text{(remainder = 1)} \end{array}$$

For the factors, just use a calculator

Now using this work, solve for ever remainder, so in this case solve for

$$r_1, r_2, r_3, 1$$

Using the lines where they're the remainder

In order, we get

$$\begin{aligned}
 r_1 &= m - k_1 \cdot a && \text{(from first)} \\
 r_2 &= a - k_2 \cdot r_1 && \text{(from second)} \\
 r_3 &= r_1 - k_3 \cdot r_2 && \text{(from third)} \\
 1 &= r_2 - k_4 \cdot r_3 && \text{(from last)}
 \end{aligned}$$

Now just work backwards and substitute each **divisor**, then combining like terms, but DO NOT actually multiply

Do this until you get to the first line with **m**. The inverse is the other factor with **a**

Example 3 (Euclidean Algorithm Practice)

Let's do some examples of the Euclidean Algorithm

Let's find the multiplicative inverse of **37** mod **200**

$$\begin{aligned}
 200 &= 5 \cdot 37 + 15 \\
 37 &= 2 \cdot 15 + 7 \\
 15 &= 2 \cdot 7 + 1
 \end{aligned}$$

Now that we found remainder = 1, let's solve for each remainder

$$\begin{aligned}
 15 &= 200 - 5 \cdot 37 \\
 7 &= 37 - 2 \cdot 15 \\
 1 &= 15 - 2 \cdot 7
 \end{aligned}$$

Now let's work backwards and substitute, starting with the bottom equation

$$\begin{aligned}
1 &= 15 - 2(37 - 2 \cdot 15) \\
1 &= 15 - 2 \cdot 37 + 4 \cdot 15 \\
1 &= 5 \cdot 15 - 2 \cdot 37 \\
1 &= 5(200 - 5 \cdot 37) - 2 \cdot 37 \\
1 &= 5 \cdot 200 - 25 \cdot 37 - 2 \cdot 37 \\
1 &= 5 \cdot 200 - 27 \cdot 37
\end{aligned}$$

From this, we see that

$$37^{-1} = -27$$

But we don't like negatives, so

$$\begin{aligned}
37^{-1} &\equiv -27 \pmod{200} \\
+0 &\equiv 200 \pmod{200}
\end{aligned}$$

$$37^{-1} \equiv 173 \pmod{200}$$

0.1.2 Greatest Common Divisor (GCD)

Let's focus our attention to a the **gcd**

Definition 4 (Greatest Common Divisor, gcd)

The Greatest Common Divisor, or gcd, gets the greatest common divisor, or factor, between two numbers

$$a, b \in \mathbb{Z}, \text{ denoted as } \gcd(a, b)$$

Such that $\gcd(a, b)$ satisfies

1. $\gcd(a, b) \mid a$ and $\gcd(a, b) \mid b$
2. $\gcd(a, b) = \max \{d \in \mathbb{Z}^+ \mid d \mid a \text{ and } d \mid b\}$

We can find the gcd by using the Euclidean Algorithm. Basically the GCD is the **last nonzero remainder**

So just do Euclidean Algorithm until the remainder is 0, don't stop at 1

Example 4 (GCD with Euclidean Algorithm)

Determine

$$\gcd(142856, 882)$$

$$142856 = 161 \cdot 882 + 854$$

$$882 = 1 \cdot 854 + 28$$

$$854 = 30 \cdot 28 + 14$$

$$28 = 2 \cdot 14 + 0$$

Since we found remainder = 0, the GCD is the last nonzero remainder, which is 14 in this case

$$\gcd(142856, 882) = 14$$

Theorem 2 (Bézout's Theorem)

Basically

$$a, b \in \mathbb{Z}^+ \implies \exists x, y \in \mathbb{Z}, ax + by = \gcd(a, b)$$

Theorem 3

If $a \mid bc$ and $\gcd(a, c) = 1$ then $a \mid b$

Proof

Since $\gcd(a, c) = 1$ then by Bézout's

$$ax + cy = \gcd(a, c) = 1$$

Multiply both sides by b and we get

$$bax + bcy = b$$

Since $a \mid bc$, that means $a \mid bcy$, and $a \mid bax$, hence $a \mid (bax + bcy)$

$$\therefore a \mid b$$

Theorem 4

If $a, b, c \in \mathbb{Z}, m \in \mathbb{Z}^+$, with $\gcd(c, m) = 1$

Then $ac \equiv bc \pmod{m}$ implies $a \equiv b \pmod{m}$

Proof

$ac \equiv bc \pmod{m}$ implies $m \mid (ac - bc)$ or $m \mid (a - b)c$

By the previous theorem

$$m \mid (a - b)$$

Therefore

$$a \equiv b \pmod{m}, \text{ as desired}$$

Note

Division by an integer $c \pmod{m}$ works if and only if $\gcd(c, m) = 1$. You may need to use Euclidean Algorithm to find c^{-1}

0.1.3 Divisors

Let's look at how to find the number of divisors in a number n

Definition 5 (Prime Factorization)

Prime factorization is breaking up a number into prime factors. An example is

$$\begin{aligned} 24 &= 6 \cdot 4 \\ &= (3 \cdot 2)(2 \cdot 2) \\ &= 3^1 \cdot 2^3 \end{aligned}$$

The prime factors are 3 and 2.

In order to find the number of divisors, we must first break up n into its prime factors

$$n = p_1^{e_1} \cdot p_2^{e_2} \cdot \dots \cdot p_k^{e_k}$$

p_k : Prime factor
 e_k : Occurrences

The number of divisors is then

$$(e_1 + 1)(e_2 + 1) \dots (e_k + 1)$$